KILNDECK · USER REQUIREMENTS

KilnDeck user requirements

Design Molecule Lab · designmolecule.com

DOCUMENT CONTROL

Document ID	KND-URS-001	Version	0.1
Status	Draft	Date	2026-10-28
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/kilndeck		
Source	docs/03a-user-requirements.md @ 9f92bd8		

REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-28	Amish Chadha	First issue (kit 1.8 upgrade), drawn from the users and stakeholders in KND-PRB-001 Tables 1a and 1b

KilnDeck user requirements

What firemen, kiln owners and fabricators need, written from their side. The product requirements that answer them are in KND-PRS-001 (`03-requirements.md`), and the links are in the generated traceability matrix KND-RTM-001 (`03b-traceability.md`). Method: STANDARDS section 22. These needs are a desk reading of the problem statement; the first co-design task (interviews with firemen) will confirm or change them.

Table 1. User requirements.

ID	Category	Title	Need	Priority	Source
UR-001	Safety	Off the crust	I feed every hole without standing or walking on the ash crust over the fire	High	Kiln fireman (PRB Table 1a)
UR-002	Safety	No fall	I cannot fall from where I work, even if I slip or lean on the rail	High	Kiln fireman
UR-003	Performance	Keep the round	Feeding a full round takes no longer than it does now, and the fuel goes into the hole, not onto the crust	High	Kiln fireman; kiln owner
UR-004	Usability	By hand	I move and feed with my own effort; there is no power on the kiln top	High	Kiln owner (PRB constraints)
UR-005	Safety	Less heat	Less heat reaches my feet and body than standing on the crust	High	Kiln fireman
UR-006	Performance	Steady dose	Each hole gets about the same amount of coal or biomass	Medium	Clean-kiln programme; kiln owner
UR-007	Deployment and assembly	Fits my kiln	It fits the kiln I have and loads only the sound walls, never the crust	High	Kiln owner
UR-008	Deployment and assembly	Two-person carry	Two people carry every piece up the kiln and bolt it together there	High	Local fabricator; kiln owner
UR-009	Logistics	Moves with the fire	It follows the fire round the kiln as it moves each day	Medium	Kiln owner
UR-010	Cost	Affordable	It costs little next to a season's firing margin	Medium	Kiln owner
UR-011	Deployment and assembly	Local build	A local workshop builds it from stock steel with hand tools, a drill and a stick welder	Medium	Local fabricator
UR-012	Environment and durability	Season-proof	It survives heat, coal dust, ash, rain and the storms of a firing season	Medium	Kiln owner
UR-013	Safety	Stays put	It cannot run away, leave its track or blow along the kiln while I am on it or while it is parked	High	Kiln fireman; kiln owner
UR-014	Maintenance	Easy upkeep	Jams can be cleared and parts greased or replaced by the firemen or a local fitter	Low	Kiln owner; local fabricator

Categories: deployment and assembly, performance, usability, safety, environment and durability, logistics, maintenance, cost, regulatory. Priorities: High, Medium, Low.